

Qatar-4b

R-2564

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_201223 · created 2026-08-26T23:25:22+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459206.6412 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1831 +/- 0.0071
Transit depth (Rp/Rs)^2	3.35 +/- 0.26 [%]
Orbital Inclination (inc)	89.85 +/- 0.98
Airmass coefficient 1 (a1)	1.055 +/- 0.033
Airmass coefficient 2 (a2)	-0.049 +/- 0.028
Scatter in the residuals of the lightcurve fit is	3.14 %
Best Comparison Star	#1 - [606, 127]
Optimal Method	PSF photometry
Transit Duration (day)	0.09571 +/- 0.00093

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.1831 ± 0.0071 vs 0.138 (deviation 5.77σ)
Duration fit vs published	0.09571 d vs 0.0892 d (deviation 6.4σ)
Mid-time O–C	6.8 min
Depth SNR	23.4
Residual scatter	3.14 %

Light curve

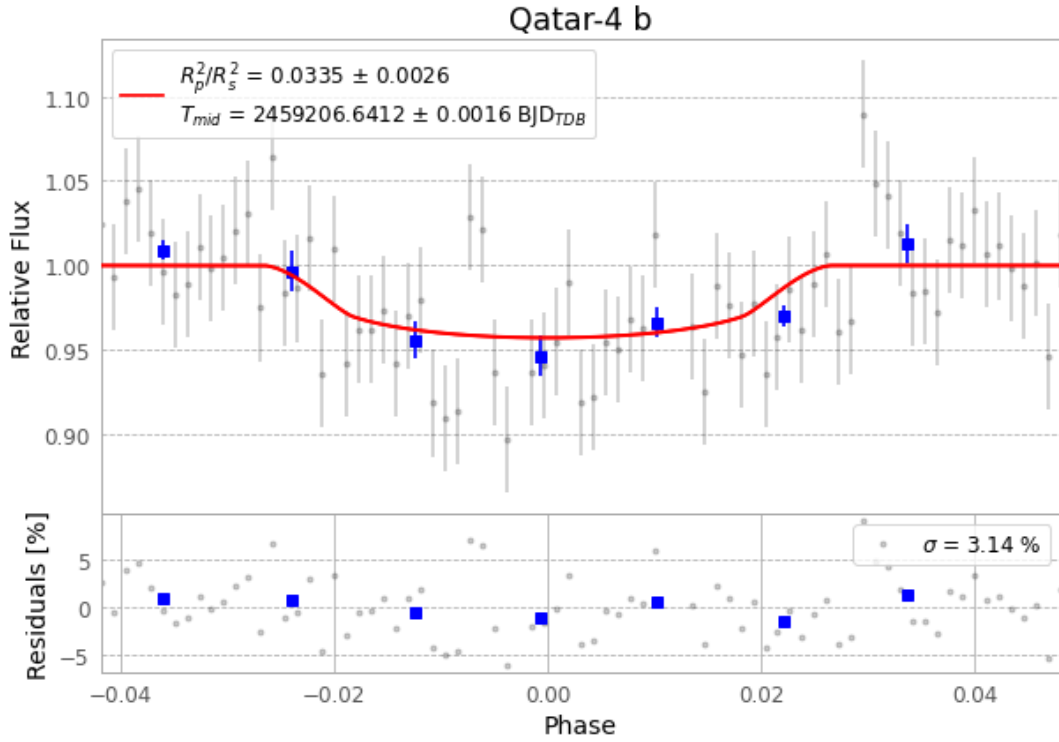


Figure 1 — FinalLightCurve_Qatar-4 b_2020-12-22.png (SHA-256 21e287c6d53a5192...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [491, 337], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5b935591f55bd8e0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

77 FITS frames (51 MB), Qatar-4201223013012.FITS → Qatar-4201223052412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-201223.log (3858dbb644bd...), FinalLightCurve_Qatar-4 b_2020-12-22.png (21e287c6d53a...), FinalLightCurve_Qatar-4 b_2020-12-22.csv (e95c34f88746...)

Compute resources

Wall time 128.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-26 23:25Z.